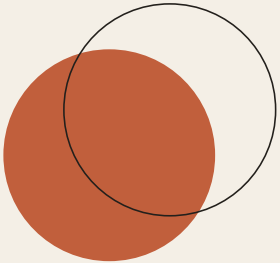
November 6, 2026

Research with AI Agents

What Works, What It Costs, and What an Institute Needs

Lukas Theissinger



Outline

I. OpenAI and Agent Usage

II. Owning Intelligence and Infrastructure at Lamarr

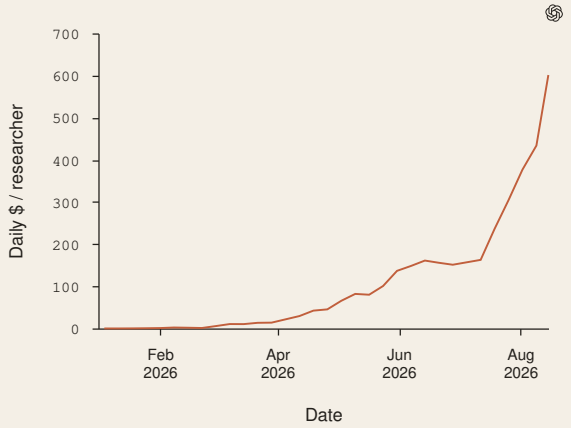


OUTLINE

I. OpenAI and Agent Usage

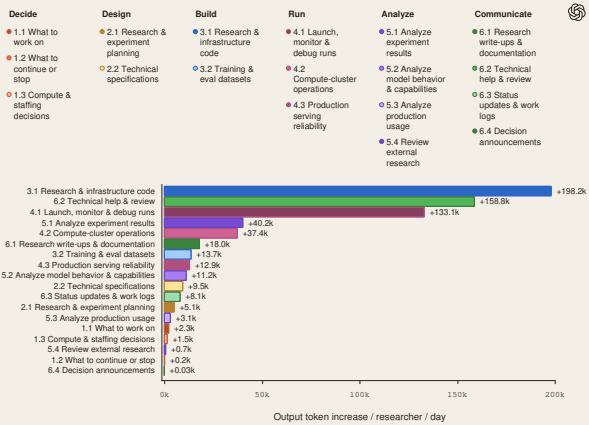
II. Owning Intelligence and Infrastructure at Lamarr

FOMO Headline: OpenAI Agent Usage



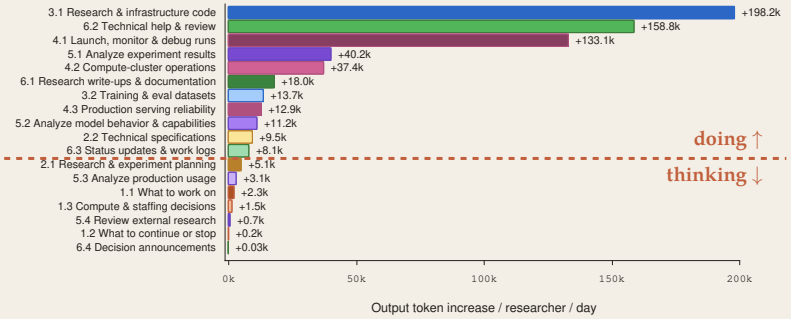
Median researcher, daily $ spent on internal coding agents. Source: OpenAI, "Research acceleration" (self-reported)

What do agents do at OpenAI?



Increase in output tokens per researcher per day, by research activity. Source: OpenAI, "Research acceleration" (self-reported)

What do agents do at OpenAI?



Increase in output tokens per researcher per day, by research activity. Source: OpenAI, "Research acceleration" (self-reported)

What Follows for Us

What the OpenAI chart suggests

All 18 activities went up: agents are used everywhere, but how much and how exactly is still open

Source: OpenAI, “Research acceleration” (self-reported)

What Follows for Us

What the OpenAI chart suggests

All 18 activities went up: agents are used everywhere, but how much and how exactly is still open

Agents took over the *doing*: code, runs, technical help

Source: OpenAI, “Research acceleration” (self-reported)

What Follows for Us

What the OpenAI chart suggests

All 18 activities went up: agents are used everywhere, but how much and how exactly is still open

Agents took over the *doing*: code, runs, technical help

Frequent, repetitive work grew most; rare decisions grew least

Source: OpenAI, “Research acceleration” (self-reported)

What Follows for Us

What the OpenAI chart suggests

All 18 activities went up: agents are used everywhere, but how much and how exactly is still open

Agents took over the *doing*: code, runs, technical help

Frequent, repetitive work grew most; rare decisions grew least

Did agents take over the *thinking*? We can't say: a plan or a decision is short, so tokens are the wrong measure

Source: OpenAI, "Research acceleration" (self-reported)

What Follows for Us

What the OpenAI chart suggests	Rules of thumb
All 18 activities went up: agents are used everywhere, but how much and how exactly is still open	Use agents or be left behind
Agents took over the *doing*: code, runs, technical help	
Frequent, repetitive work grew most; rare decisions grew least	
Did agents take over the *thinking*? We can't say: a plan or a decision is short, so tokens are the wrong measure	

Source: OpenAI, "Research acceleration" (self-reported)
Source: Tim Dettmers (CMU; author of QLoRA and bitsandbytes), "Use Agents or Be Left Behind?", Jan 2026, timdettmers.com

What Follows for Us

What the OpenAI chart suggests	Rules of thumb
All 18 activities went up: agents are used everywhere, but how much and how exactly is still open	Use agents or be left behind
Agents took over the *doing*: code, runs, technical help	More than 90% of code and text should be written by agents.
Frequent, repetitive work grew most; rare decisions grew least	
Did agents take over the *thinking*? We can't say: a plan or a decision is short, so tokens are the wrong measure	

Source: OpenAI, "Research acceleration" (self-reported)
Source: Tim Dettmers (CMU; author of QLoRA and bitsandbytes), "Use Agents or Be Left Behind?", Jan 2026, timdettmers.com

What Follows for Us

What the OpenAI chart suggests	Rules of thumb
All 18 activities went up: agents are used everywhere, but how much and how exactly is still open	Use agents or be left behind
Agents took over the *doing*: code, runs, technical help	More than 90% of code and text should be written by agents.
Frequent, repetitive work grew most; rare decisions grew least	Automate only if how often × time saved > time to build
Did agents take over the *thinking*? We can't say: a plan or a decision is short, so tokens are the wrong measure	

Source: OpenAI, "Research acceleration" (self-reported)
Source: Tim Dettmers (CMU; author of QLoRA and bitsandbytes), "Use Agents or Be Left Behind?", Jan 2026, timdettmers.com

What Follows for Us

What the OpenAI chart suggests	Rules of thumb
All 18 activities went up: agents are used everywhere, but how much and how exactly is still open	Use agents or be left behind
Agents took over the *doing*: code, runs, technical help	More than 90% of code and text should be written by agents.
Frequent, repetitive work grew most; rare decisions grew least	Automate only if how often $\times$ time saved $>$ time to build
Did agents take over the *thinking*? We can't say: a plan or a decision is short, so tokens are the wrong measure	Agent use is a *skill*: not everything can or should be automated (degree of automation)

Source: OpenAI, "Research acceleration" (self-reported)
Source: Tim Dettmers (CMU; author of QLoRA and bitsandbytes), "Use Agents or Be Left Behind?", Jan 2026, timdettmers.com

Field Report: Automating Exercises

1. **The problem.** PhD student, tutoring an exercise class every two weeks:
   - 2 h exercise class + 6 h preparation (solving the tasks, making the slides)
   - rough estimate: **2 h + 6 h = 8 h every two weeks**

Field Report: Automating Exercises

1. **The problem.** PhD student, tutoring an exercise class every two weeks:
   - 2 h exercise class + 6 h preparation (solving the tasks, making the slides)
   - rough estimate: **2 h + 6 h = 8 h every two weeks**

2. **Automate as much as possible**
   a) Task solving: an agent solves every task, with code and plots
   b) Slide creation: an agent turns the solutions into the class slides

Field Report: Automating Exercises

1. **The problem.** PhD student, tutoring an exercise class every two weeks:
   - 2 h exercise class + 6 h preparation (solving the tasks, making the slides)
   - rough estimate: **2 h + 6 h = 8 h every two weeks**

2. **Automate as much as possible**
   a) Task solving: an agent solves every task, with code and plots
   b) Slide creation: an agent turns the solutions into the class slides

3. **Left to do:** understand the solutions, make small updates to the slides
   - class stays 2 h, preparation drops from 6 h to 2 h: **2 h + 2 h = 4 h every two weeks**

Field Report: Automating Exercises

1. **The problem.** PhD student, tutoring an exercise class every two weeks:
   - 2 h exercise class + 6 h preparation (solving the tasks, making the slides)
   - rough estimate: **2 h + 6 h = 8 h every two weeks**

2. **Automate as much as possible**
   a) Task solving: an agent solves every task, with code and plots
   b) Slide creation: an agent turns the solutions into the class slides

3. **Left to do:** understand the solutions, make small updates to the slides
   - class stays 2 h, preparation drops from 6 h to 2 h: **2 h + 2 h = 4 h every two weeks**

4. **Ask your agent to try it:** `github.com/Newida/AutomatingExercises` **Avoid:** Don't do anything BY HAND on this project. If you want to change it, ASK YOUR AGENT to change it

II

OUTLINE

I. OpenAI and Agent Usage

II. Owning Intelligence and Infrastructure at Lamarr

Automating at Bigger Scale

- **So far:** each of us automates our own tasks

Automating at Bigger Scale

- **So far:** each of us automates our own tasks
- **Share it:** one person builds it, everyone reuses and adapts it to their needs
  We need a place our agents can look for shared Lamarr automatizations

Automating at Bigger Scale

- **So far:** each of us automates our own tasks
- **Share it:** one person builds it, everyone reuses and adapts it to their needs
  We need a place our agents can look for shared Lamarr automatizations
- **Institute level:** tasks that never pay off for one person start to pay off

Automating at Bigger Scale

- **So far:** each of us automates our own tasks
- **Share it:** one person builds it, everyone reuses and adapts it to their needs
  We need a place our agents can look for shared Lamarr automatizations
- **Institute level:** tasks that never pay off for one person start to pay off

Example: travel expense reports

Collect receipts, fill in the university form, look up per-diem rules, fix what admin sends back: say **3 trips** × **3 h per person per year**. An agent that reads the receipts and files the form correctly needs ∼**200 h** (rules, forms, access to the admin system)

$$\text{how often} \times \text{time saved} \qquad \text{time to build}$$

Automating at Bigger Scale

- **So far:** each of us automates our own tasks
- **Share it:** one person builds it, everyone reuses and adapts it to their needs
  We need a place our agents can look for shared Lamarr automatizations
- **Institute level:** tasks that never pay off for one person start to pay off

EXAMPLE: TRAVEL EXPENSE REPORTS

Collect receipts, fill in the university form, look up per-diem rules, fix what admin sends back: say **3 trips** × **3 h per person per year**. An agent that reads the receipts and files the form correctly needs ∼**200 h** (rules, forms, access to the admin system)

	how often × time saved	time to build	
one person	$3 \times 3\,\text{h} = 9\,\text{h/year}$	200 h	**never pays**

Automating at Bigger Scale

- **So far:** each of us automates our own tasks
- **Share it:** one person builds it, everyone reuses and adapts it to their needs
  We need a place our agents can look for shared Lamarr automatizations
- **Institute level:** tasks that never pay off for one person start to pay off

EXAMPLE: TRAVEL EXPENSE REPORTS

Collect receipts, fill in the university form, look up per-diem rules, fix what admin sends back: say **3 trips** × **3 h per person per year**. An agent that reads the receipts and files the form correctly needs ∼**200 h** (rules, forms, access to the admin system)

	how often × time saved	time to build	
one person	$3 \times 3\,\text{h} = 9\,\text{h/year}$	200 h	**never pays**
institute, 10 000 people	$10\,000 \times 9\,\text{h} = 90\,000\,\text{h/year}$	200 h	**pays >400×**

What Agents Cost

- **Me:** Claude *and* Codex, 20 € plan each: **40 € per month**. One plan alone is not enough

What Agents Cost

- **Me:** Claude *and* Codex, 20 € plan each: **40 € per month**. One plan alone is not enough
- **Colleagues:** 100 € Claude plan, and they hit its limit **every week**

What Agents Cost

- **Me:** Claude *and* Codex, 20 € plan each: **40 € per month**. One plan alone is not enough
- **Colleagues:** 100 € Claude plan, and they hit its limit **every week**
- **As we have seen:** every Lamarr member needs AI agents. You cannot work without them anymore

What Agents Cost

- **Me:** Claude *and* Codex, 20 € plan each: **40 € per month**. One plan alone is not enough
- **Colleagues:** 100 € Claude plan, and they hit its limit **every week**
- **As we have seen:** every Lamarr member needs AI agents. You cannot work without them anymore

A request to management: right now, many of us pay for these tools ourselves. Most of us stop at the 20 € plan rather than the 100 € one, so we get less out of the tools than we could. Institutional access would help us do better work and it will matter more and more for attracting and keeping good people, as other institutes, in Germany and abroad, start to provide it

What Agents Cost

- **Me:** Claude *and* Codex, 20 € plan each: **40 € per month**. One plan alone is not enough
- **Colleagues:** 100 € Claude plan, and they hit its limit **every week**
- **As we have seen:** every Lamarr member needs AI agents. You cannot work without them anymore

A request to management: right now, many of us pay for these tools ourselves. Most of us stop at the 20 € plan rather than the 100 € one, so we get less out of the tools than we could. Institutional access would help us do better work and it will matter more and more for attracting and keeping good people, as other institutes, in Germany and abroad, start to provide it

So let's talk about costs

Decision 1: Subscription or API?

Subscription (20–200 €/month)

- \+ flat price, no surprise bills
- \+ very cheap for heavy interactive use
- – usage caps (5-hour and weekly)
- – one person, one login; not for pipelines

Decision 1: Subscription or API?

Subscription (20–200 €/month)

- \+ flat price, no surprise bills
- \+ very cheap for heavy interactive use
- – usage caps (5-hour and weekly)
- – one person, one login; not for pipelines

API (pay per token)

- \+ no caps, run many agents in parallel
- \+ scripts, batch jobs, shared tools
- – heavy daily use gets expensive fast
- – needs a budget and someone watching it

Decision 1: Subscription or API?

Subscription (20–200 €/month)

- \+ flat price, no surprise bills
- \+ very cheap for heavy interactive use
- – usage caps (5-hour and weekly)
- – one person, one login; not for pipelines

API (pay per token)

- \+ no caps, run many agents in parallel
- \+ scripts, batch jobs, shared tools
- – heavy daily use gets expensive fast
- – needs a budget and someone watching it

One month of my usage, priced at API list prices

	subscription	same usage via API	factor
Claude	20 $ (Pro)	∼680 $	∼34×
Codex	20 $ (Plus)	∼145 $	∼7×

Decision 2: Self-Hosting? A Field Report

Main model: GLM-5.3-Flash, 45 days

	45 days	per month
input tokens	28.3 B	∼19 B
of it cached	25.4 B (90 %)	
output tokens	585 M	∼390 M
at Z.ai prices	$4.5 k	∼$3 k
in Sonnet prices	$14–25 k	∼$9–17 k
energy (full load)	500–1000 €	∼330–670 €
hardware	**[?]**	**[?]**
staff time	**[?]**	**[h/week?]**

Decision 2: Self-Hosting? A Field Report

Main model: GLM-5.3-Flash, 45 days

	45 days	per month
input tokens	28.3 B	∼19 B
of it cached	25.4 B (90 %)	
output tokens	585 M	∼390 M
at Z.ai prices	$4.5 k	∼$3 k
in Sonnet prices	$14–25 k	∼$9–17 k
energy (full load)	500–1000 €	∼330–670 €
hardware	[?]	[?]
staff time	[?]	[h/week?]

- **Also:** Qwen 3.8 27B, DeepSeek V4.1 Flash, GLM 5.3, … rotated weekly: >$15 k at API prices (≈$45–75 k in Sonnet prices)

Decision 2: Self-Hosting? A Field Report

Main model: GLM-5.3-Flash, 45 days

	45 days	per month
input tokens	28.3 B	∼19 B
of it cached	25.4 B (90 %)	
output tokens	585 M	∼390 M
at Z.ai prices	$4.5 k	∼$3 k
in Sonnet prices	$14–25 k	∼$9–17 k
energy (full load)	500–1000 €	∼330–670 €
hardware	[?]	[?]
staff time	[?]	[h/week?]

- **Also:** Qwen 3.8 27B, DeepSeek V4.1 Flash, GLM 5.3, … rotated weekly: >$15 k at API prices (≈$45–75 k in Sonnet prices)
- **What made it work:** aggressive caching (incl. 256 GB CPU cache), a proxy that puts interactive coding ahead of bulk jobs

Decision 2: Self-Hosting? A Field Report

Main model: GLM-5.3-Flash, 45 days

	45 days	per month
input tokens	28.3 B	∼19 B
of it cached	25.4 B (90 %)	
output tokens	585 M	∼390 M
at Z.ai prices	$4.5 k	∼$3 k
in Sonnet prices	$14–25 k	∼$9–17 k
energy (full load)	500–1000 €	∼330–670 €
hardware	[?]	[?]
staff time	[?]	[h/week?]

- **Also:** Qwen 3.8 27B, DeepSeek V4.1 Flash, GLM 5.3, … rotated weekly: >$15 k at API prices (≈$45–75 k in Sonnet prices)
- **What made it work:** aggressive caching (incl. 256 GB CPU cache), a proxy that puts interactive coding ahead of bulk jobs
- **The catch:** he had to patch vLLM himself, the difference between "completely unusable" and "fairly usable"

Decision 2: Self-Hosting? A Field Report

Main model: GLM-5.3-Flash, 45 days

	45 days	per month
input tokens	28.3 B	∼19 B
of it cached	25.4 B (90 %)	
output tokens	585 M	∼390 M
at Z.ai prices	$4.5 k	∼$3 k
in Sonnet prices	$14–25 k	∼$9–17 k
energy (full load)	500–1000 €	∼330–670 €
hardware	[?]	[?]
staff time	[?]	[h/week?]

- **Also:** Qwen 3.8 27B, DeepSeek V4.1 Flash, GLM 5.3, … rotated weekly: >$15 k at API prices (≈$45–75 k in Sonnet prices)
- **What made it work:** aggressive caching (incl. 256 GB CPU cache), a proxy that puts interactive coding ahead of bulk jobs
- **The catch:** he had to patch vLLM himself, the difference between "completely unusable" and "fairly usable"
- **Quality:** [vs. Claude/GPT on agentic coding? agent tool?]
- **Speed:** [tokens/s per user, at peak?]
- **Utilization:** [GPUs busy on average? idle at night?]

Decision 2: Self-Hosting? A Field Report

Main model: GLM-5.3-Flash, 45 days

	45 days	per month
input tokens	28.3 B	∼19 B
of it cached	25.4 B (90 %)	
output tokens	585 M	∼390 M
at Z.ai prices	$4.5 k	∼$3 k
in Sonnet prices	$14–25 k	∼$9–17 k
energy (full load)	500–1000 €	∼330–670 €
hardware	[?]	[?]
staff time	[?]	[h/week?]

- **Also:** Qwen 3.8 27B, DeepSeek V4.1 Flash, GLM 5.3, … rotated weekly: >$15 k at API prices (≈$45–75 k in Sonnet prices)
- **What made it work:** aggressive caching (incl. 256 GB CPU cache), a proxy that puts interactive coding ahead of bulk jobs
- **The catch:** he had to patch vLLM himself, the difference between "completely unusable" and "fairly usable"
- **Quality:** [vs. Claude/GPT on agentic coding? agent tool?]
- **Speed:** [tokens/s per user, at peak?]
- **Utilization:** [GPUs busy on average? idle at night?]

Takeaway

Running the model is cheap (energy ≪ API bill). The real costs are the hardware [?] and the expert who keeps it running

Takeaways

For the individual

1. **You need to use AI agents**

Takeaways

For the individual

1. **You need to use AI agents**
2. **Use AI agents to change your bottlenecks at work**

Takeaways

For the individual

1. **You need to use AI agents**
2. **Use AI agents to change your bottlenecks at work**
3. **Automate, to a degree, only what pays off:**

   how often $\times$ time saved $>$ time to build

Takeaways

For the individual

1. **You need to use AI agents**
2. **Use AI agents to change your bottlenecks at work**
3. **Automate, to a degree, only what pays off:**
   how often $\times$ time saved $>$ time to build
4. **Use AI to build your own AI setup**
   Learn by doing, tailor it to your workflow, and make your daily work easier

Takeaways

For the individual

1. **You need to use AI agents**
2. **Use AI agents to change your bottlenecks at work**
3. **Automate, to a degree, only what pays off:**
   how often × time saved > time to build
4. **Use AI to build your own AI setup**
   Learn by doing, tailor it to your workflow, and make your daily work easier

For the institute

1. **Automate workflows for every individual:** what never pays off for one person pays off for 10 000

Takeaways

For the individual

1. **You need to use AI agents**
2. **Use AI agents to change your bottlenecks at work**
3. **Automate, to a degree, only what pays off:**
   how often × time saved > time to build
4. **Use AI to build your own AI setup**
   Learn by doing, tailor it to your workflow, and make your daily work easier

For the institute

1. **Automate workflows for every individual:** what never pays off for one person pays off for 10 000
2. **Provide your employees with agents**

Takeaways

For the individual

1. **You need to use AI agents**
2. **Use AI agents to change your bottlenecks at work**
3. **Automate, to a degree, only what pays off:**
   how often × time saved > time to build
4. **Use AI to build your own AI setup**
   Learn by doing, tailor it to your workflow, and make your daily work easier

For the institute

1. **Automate workflows for every individual:** what never pays off for one person pays off for 10 000
2. **Provide your employees with agents**
3. **Consider self-hosting**

Next: David's Setup

1. We know that we **need AI agents** and what to use them for

Next: David's Setup

1. We know that we **need AI agents** and what to use them for
2. David shows **his own setup**: how it looks in daily work for him

Next: David's Setup

1. We know that we **need AI agents** and what to use them for
2. David shows **his own setup**: how it looks in daily work for him
3. The best setup is **personal and custom**, so build your own